

Chapter 1

Introduction to Data Mining

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Contents

- Why data mining?
- What is data mining?
- Data Mining Strategies
- CRISP-DM Model
- What kind of data for mining?
- Data mining functionality
- Data mining applications

Why Data Mining?

- Data explosion problem

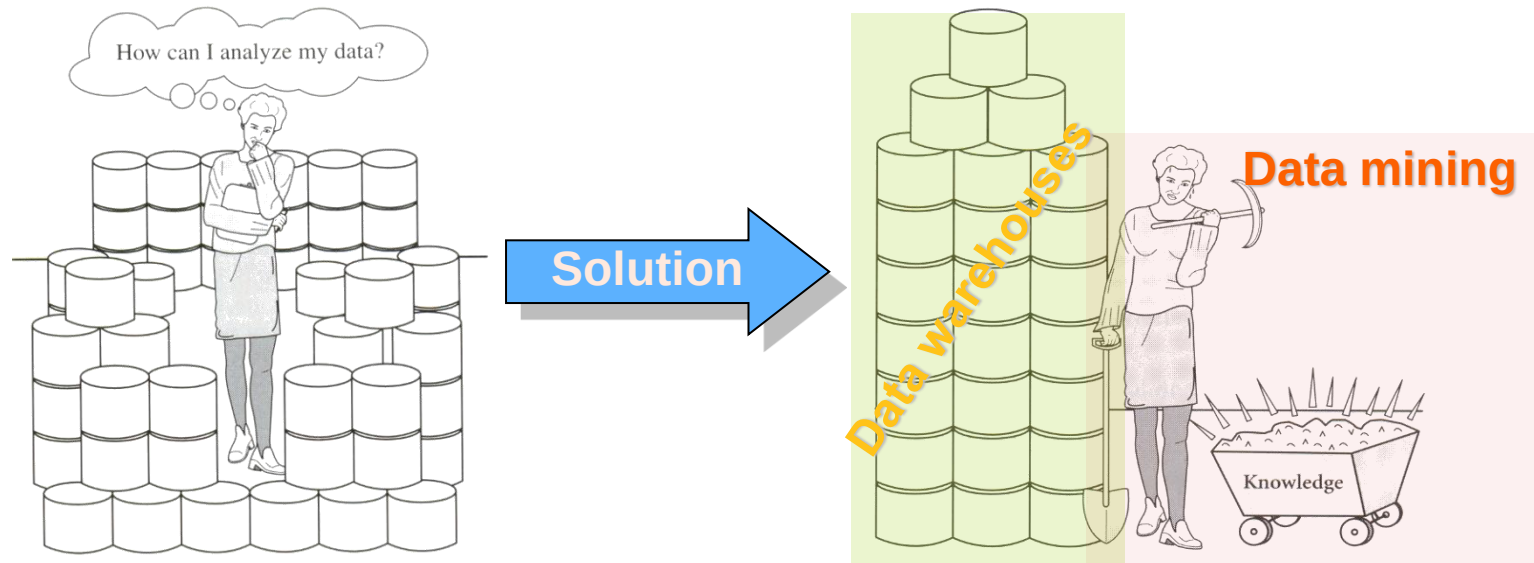


**We are drowning in data,
but starving for knowledge!**

- Major sources of abundant data
 - **Business**: Web, e-commerce, transactions, stocks, ...
 - **Science**: Remote sensing, bioinformatics
 - **Society and everyone**: YouTube, Facebook, Twitter

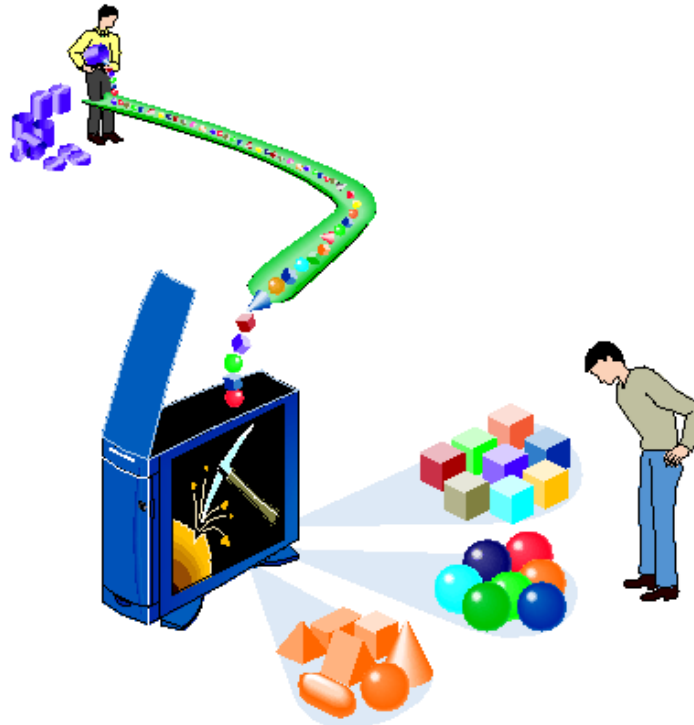
Why Data Mining?

- Solution: Data warehousing and data mining
 - Data warehousing and on-line analytical processing
 - Extraction of interesting knowledge (rules, regularities, patterns, constraints) from data in large databases

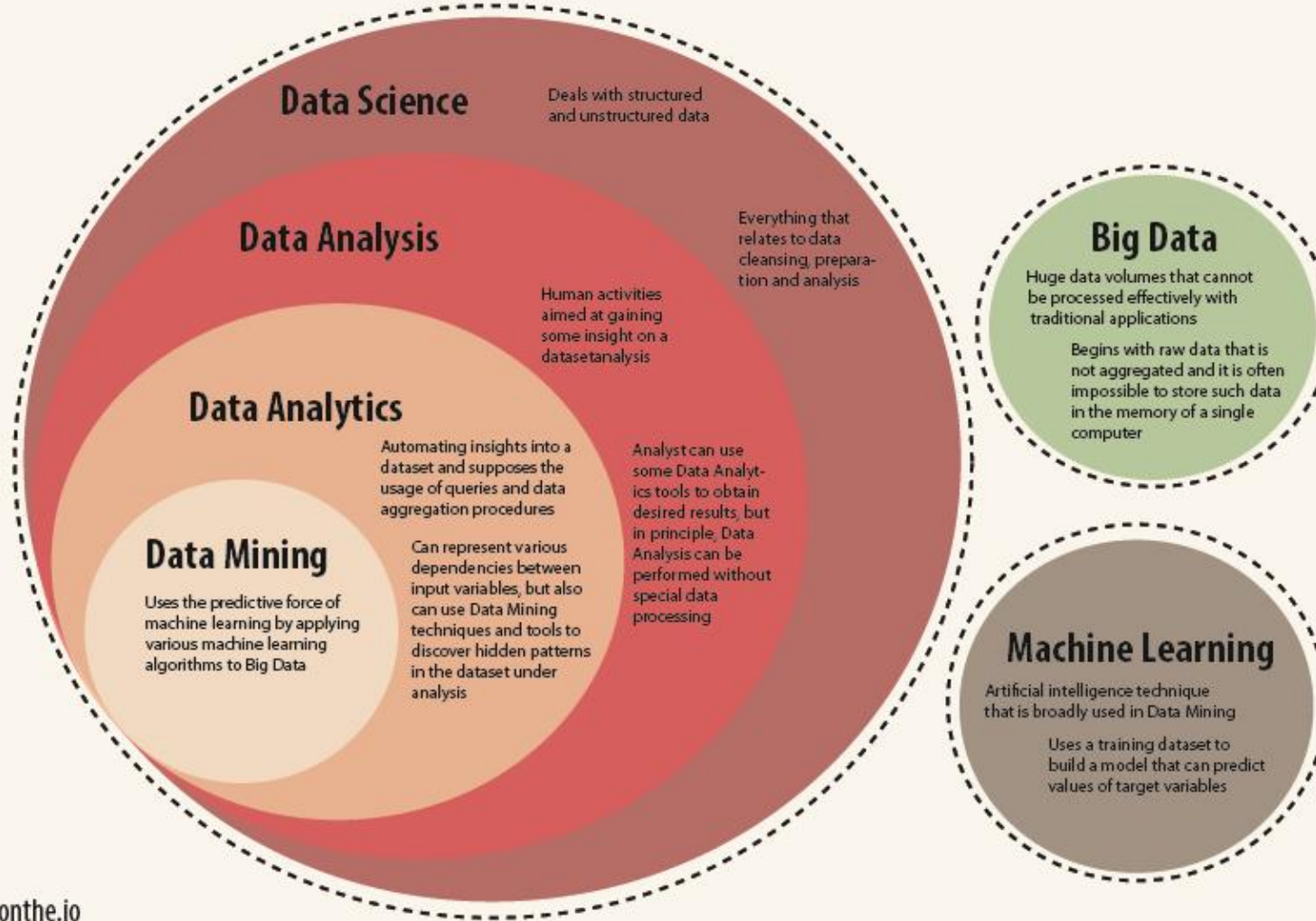


What is Data Mining?

- Data mining
 - Extraction of interesting information or patterns from data in **large databases**



What is the difference between Data Science, Data Analysis, Big Data, Data Analytics, Data Mining and Machine Learning?

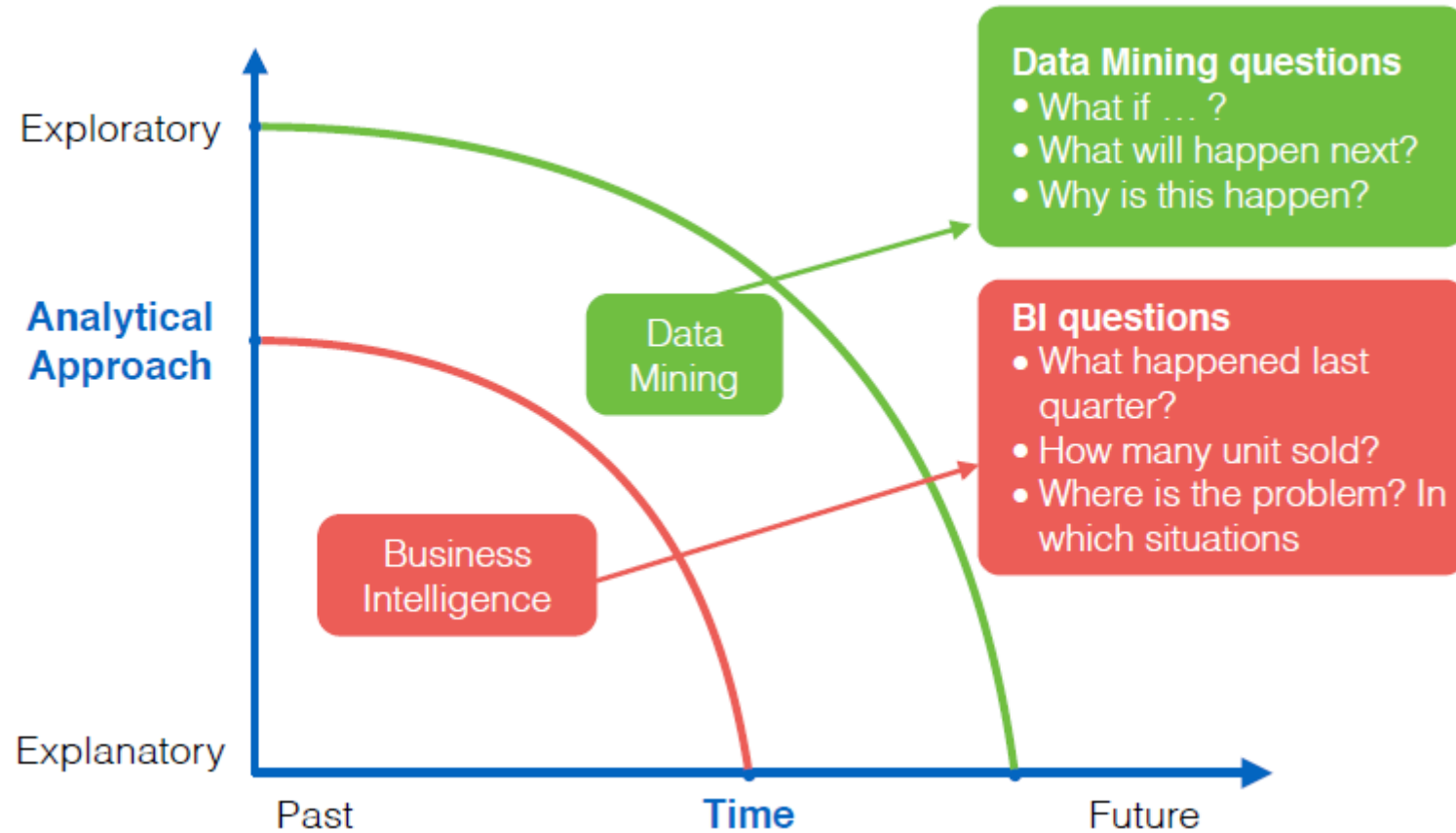


Source: onthe.io

What is Data Mining?

- Alternative names and their “inside stories”:
 - Data mining: a misnomer?
 - Knowledge Discovery in Databases (KDD), knowledge extraction, data/pattern analysis, business intelligence, etc.
- What is not data mining?
 - Simple search and query processing.
 - Expert systems (Deductive)
or small ML/statistical programs

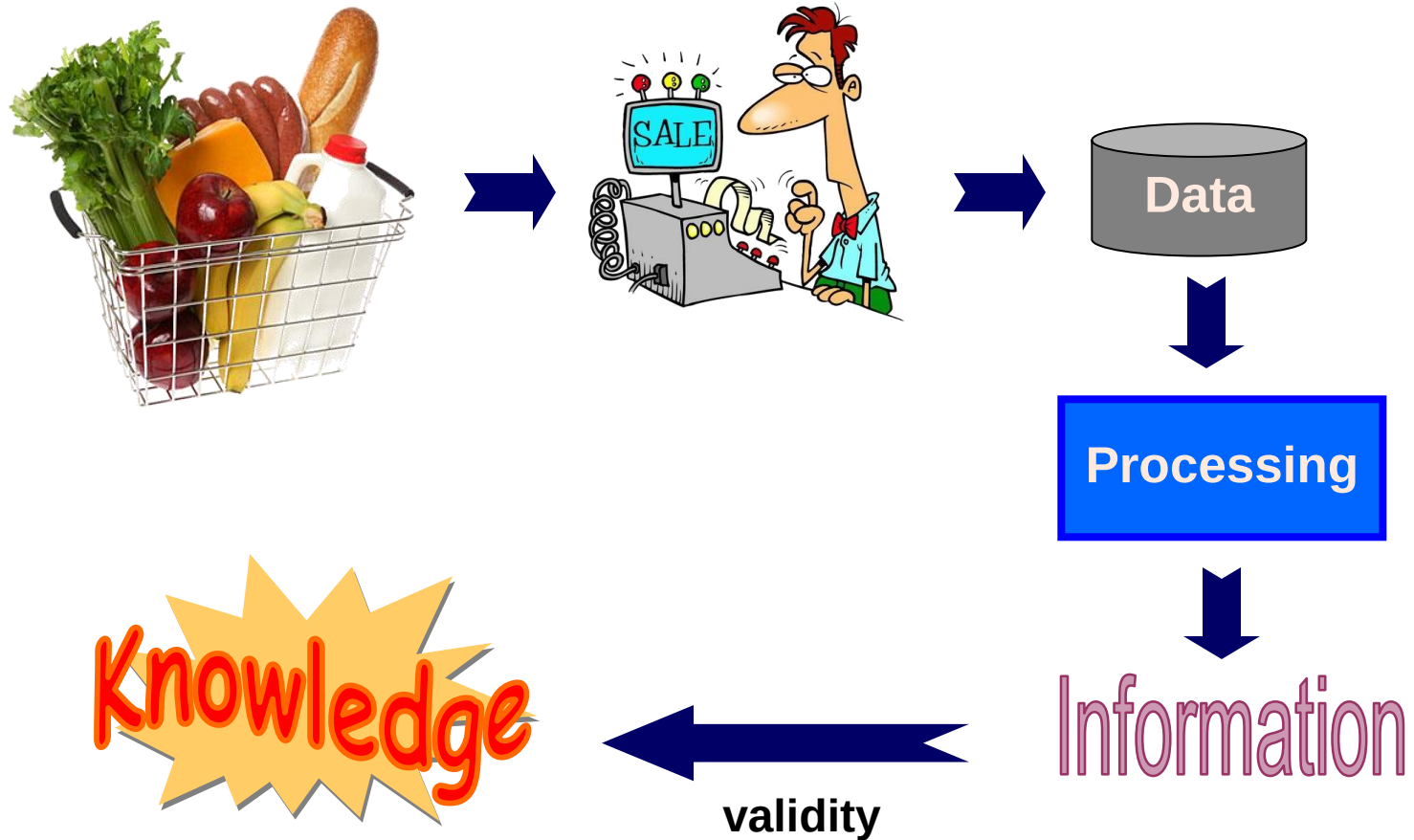
Business Intelligence vs Data Mining



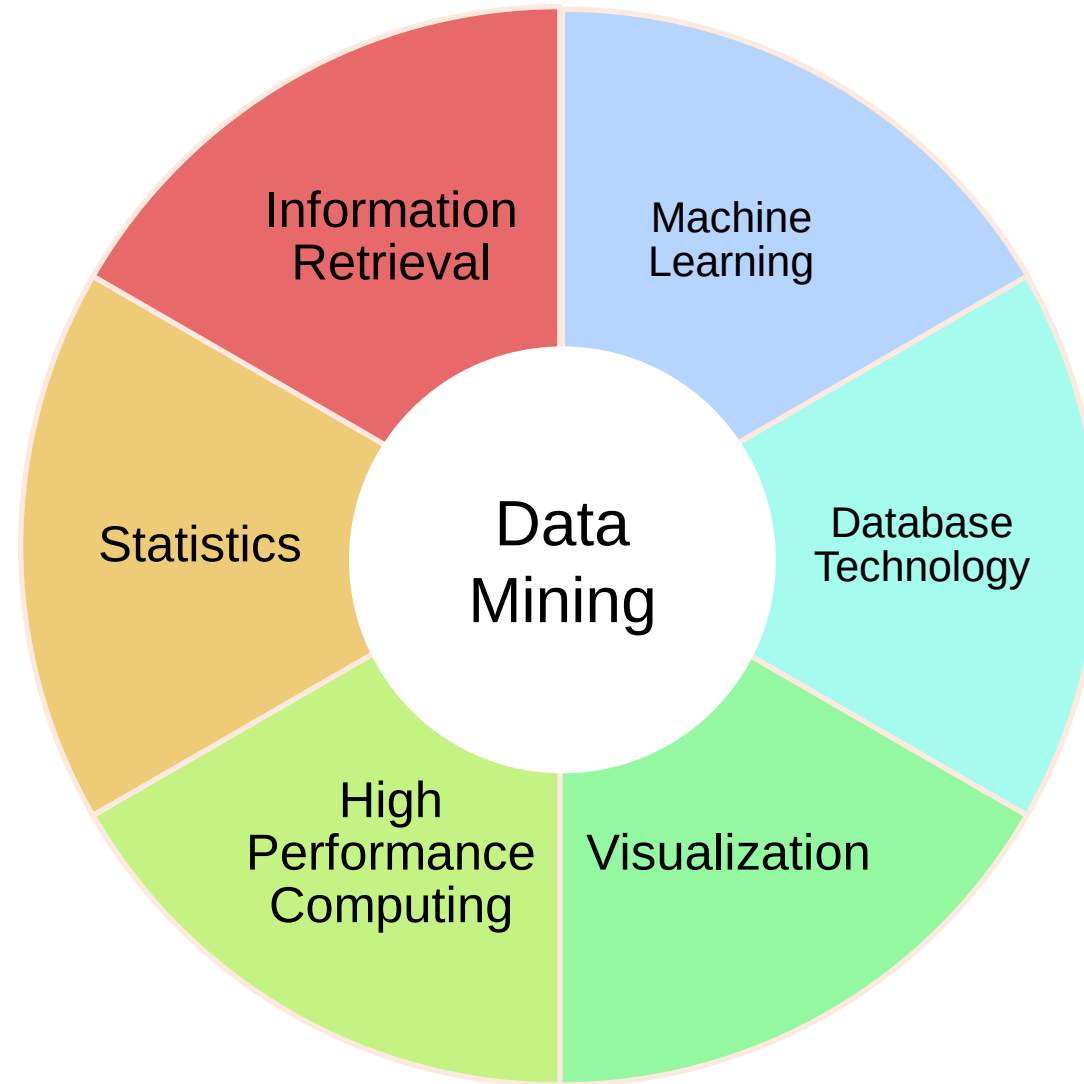
Characteristics of Knowledge

- Nontrivial
- Valid
- Novel / Previously unknown
- Useful
- Interesting
- Understandable

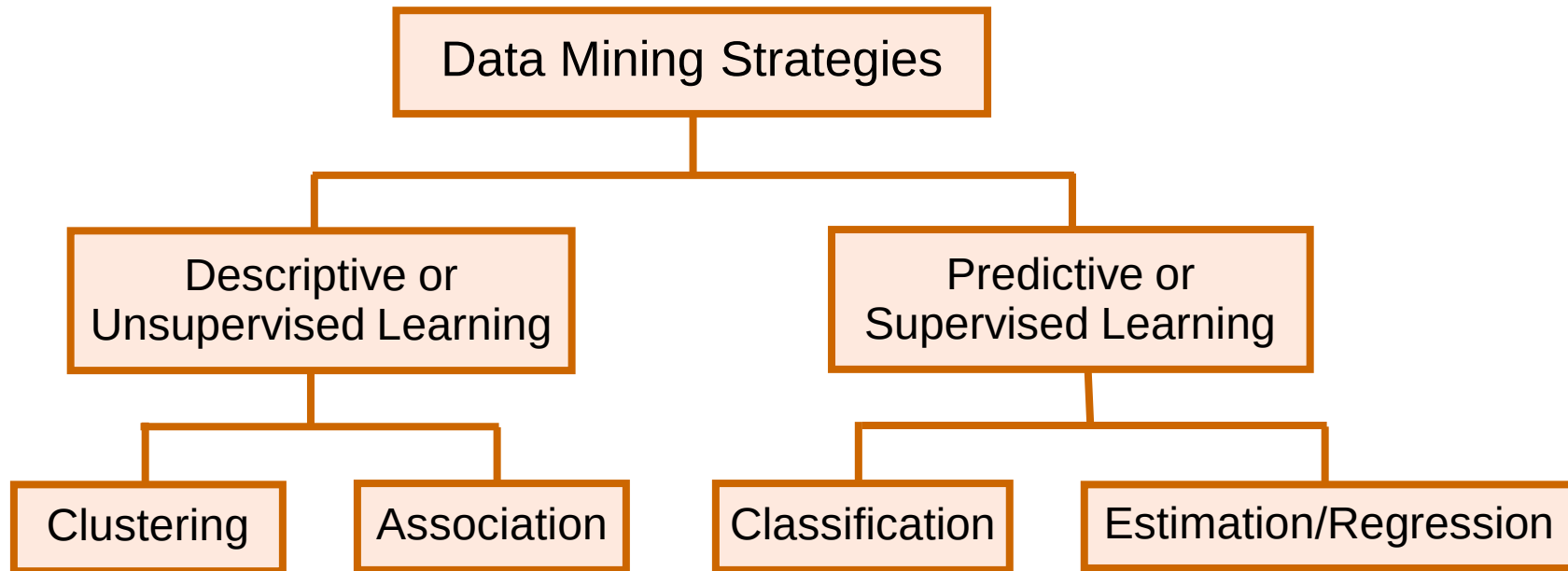
Data Information and Knowledge



Data Mining: Confluence of Multiple Disciplines



Data Mining Strategies



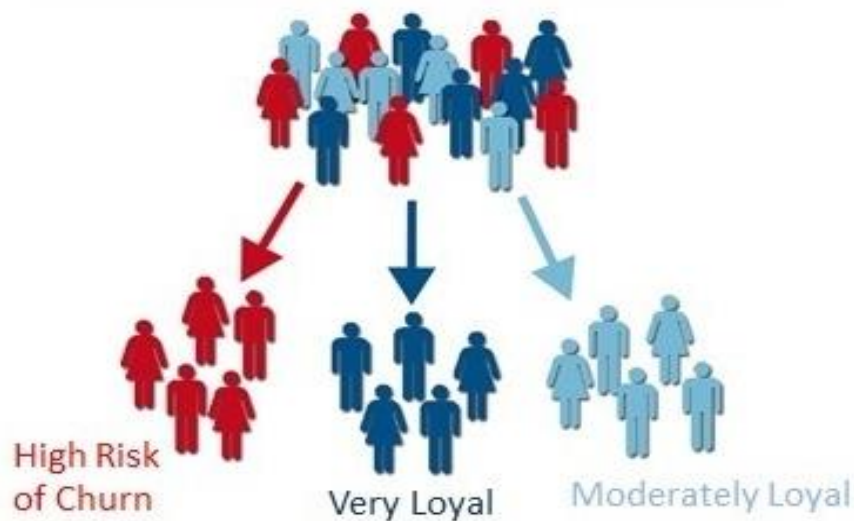
Data Mining Strategies

- Supervised Learning
 - Process of building classification models using data instances of known origin to learning with predefined classes
 - Build models by using input attributes (independent variables) to predict output or dependent variables (class label)
- Unsupervised Learning
 - To discover natural grouping or concept structures in data
 - Without a dependent variable to guide learning process
 - Rather, it builds knowledge structures by using some measures of cluster quality to group instances into classes

Supervised Learning Models

	Purpose	Output
Classification	Determine a value for an unknown output attribute (deal with current behavior)	Categorical (discrete)
Estimation		Numerical (continuous)

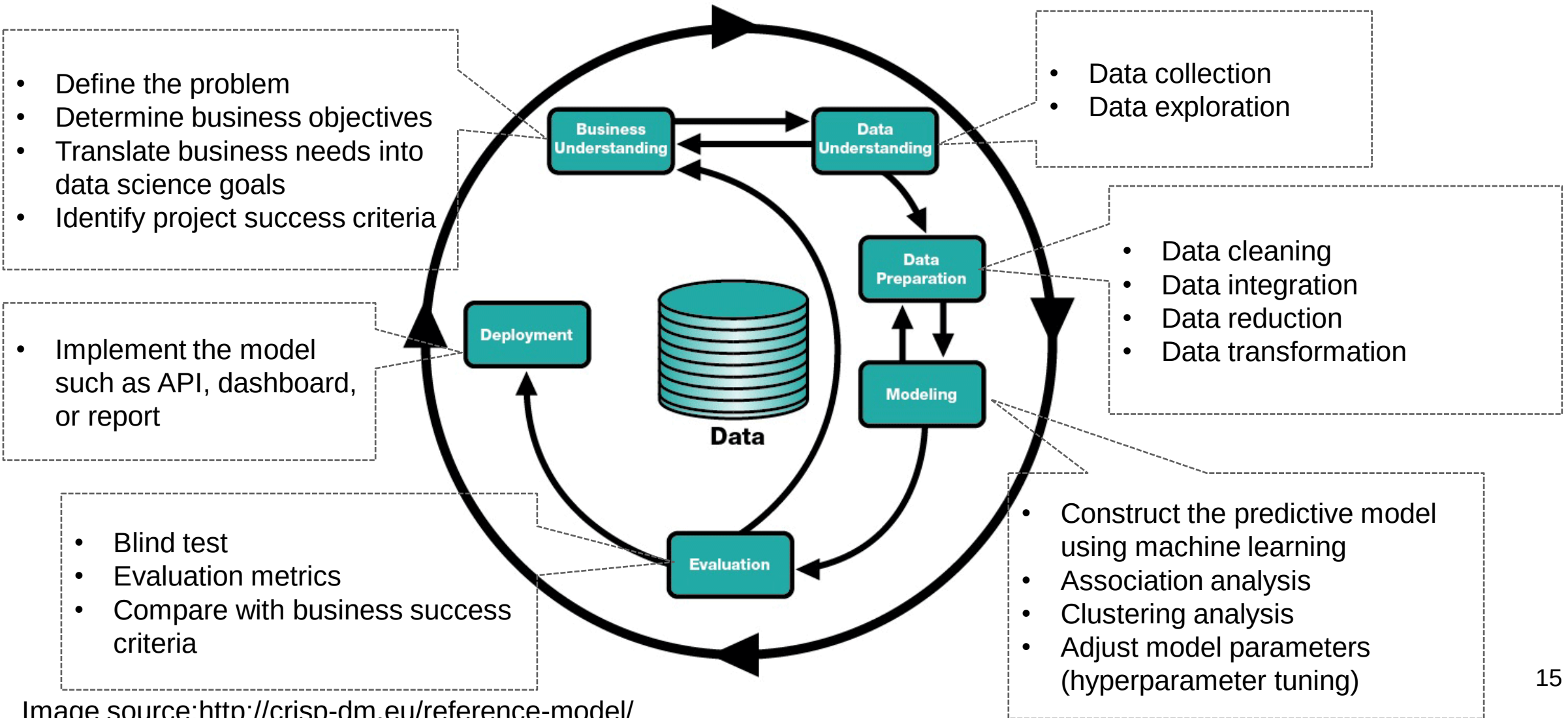
classification



regression

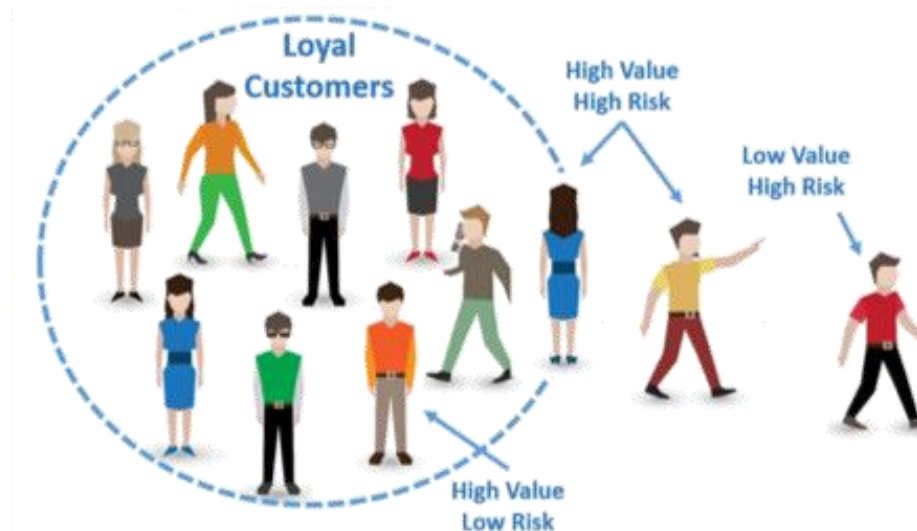


- CRISP-DM Model: **C**ross **I**ndustry **S**tandard **P**rocess for **D**ata **M**ining



CRISP-DM: Churn Prediction

- Business Understanding
 - Understand why predicting churn matters and what the business wants to achieve
 - Define KPI such as churn rate, retention rate
 - Business objective → detecting which customers are likely to leave a service or to cancel a subscription to a service
 - Data mining goal → classification model
 - Predict whether a customer will churn in the next 30 days to reduce churn rate from 15% to 10%

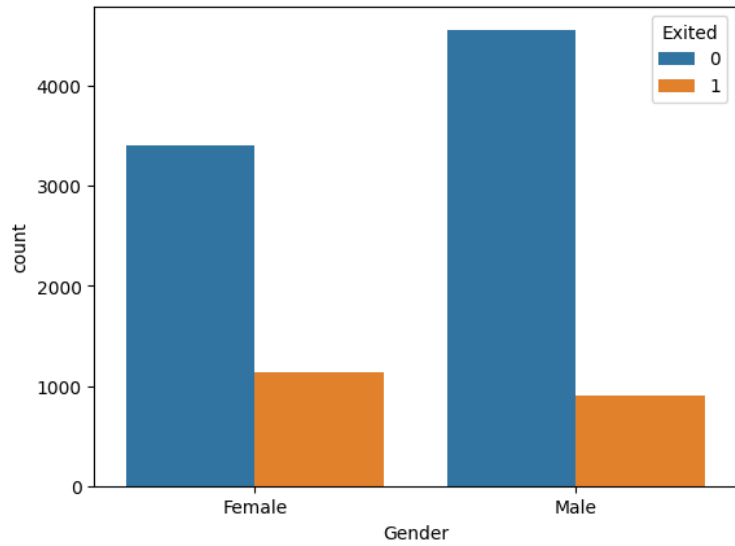
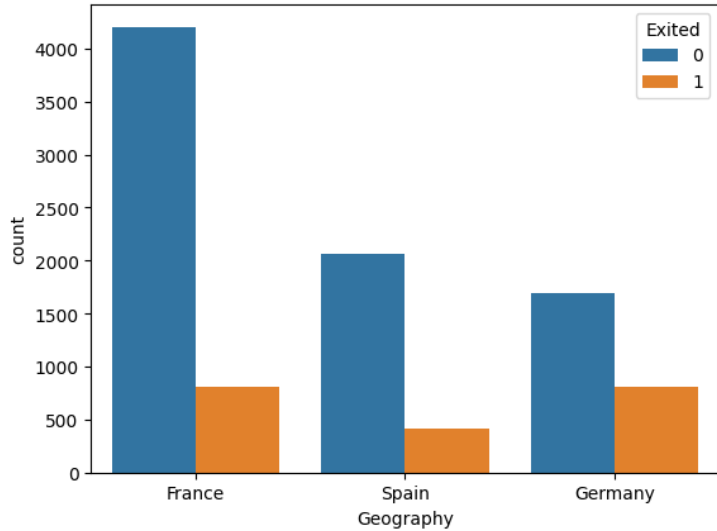
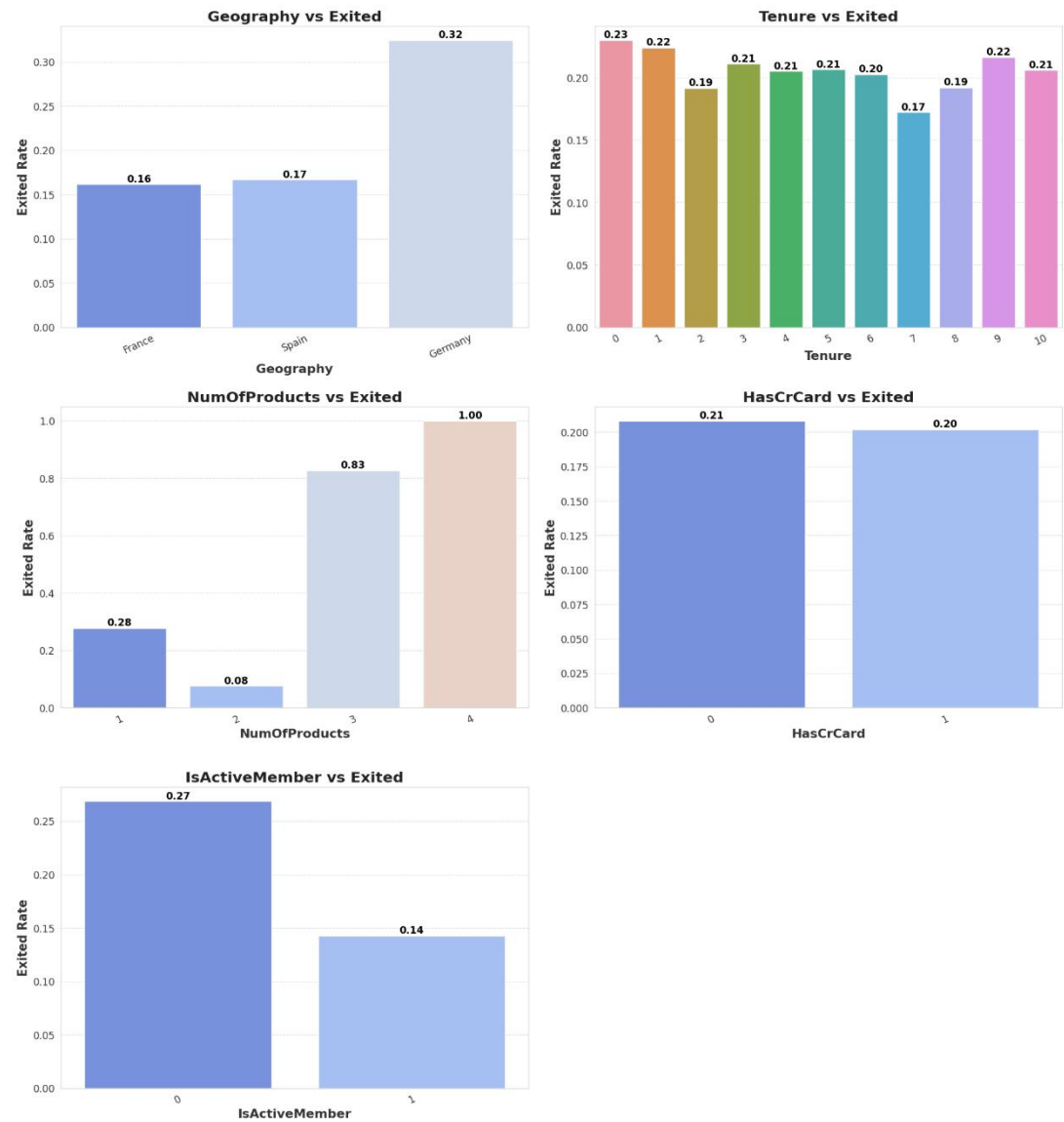


CRISP-DM: Churn Prediction

- Data Understanding
 - Collect data, explore and identify quality of data
 - Data Exploration (14 columns and 10,000 rows)

Row Number	CustomerId	Surname	Credit Score	Geography	Gender	Age	Tenure	Balance	NumOf Product	HasCr Card	IsActive Member	Estimated Salary	Exited
1	15634602	Hargrave	619	France	Female	42	2	0	1	1	1	101348.9	1
2	15647311	Hill	608	Spain	Female	41	1	83807.86	1	0	1	112542.6	0
3	15619304	Onio	502	France	Female	42	8	159660.8	3	1	0	113931.6	1
4	15701354	Boni	699	France	Female	39	1	0	2	0	0	93826.63	0
5	15737888	Mitchell	850	Spain	Female	43	2	125510.8	1	1	1	79084.1	0
6	15574012	Chu	645	Spain	Male	44	8	113755.8	2	1	0	149756.7	1
7	15592531	Bartlett	822	France	Male	50	7	0	2	1	1	10062.8	0
8	15656148	Obinna	376	Germany	Female	29	4	115046.7	4	1	0	119346.9	1
9	15792365	He	501	France	Male	44	4	142051.1	2	0	1	74940.5	0
10	15592389	H?	684	France	Male	27	2	134603.9	1	1	1	71725.73	0
11	15767821	Bearce	528	France	Male	31	6	102016.7	2	0	0	80181.12	0
12	15737173	Andrews	497	Spain	Male	24	3	0	2	1	0	76390.01	0
13	15632264	Kay	476	France	Female	34	10	0	2	1	0	26260.98	0

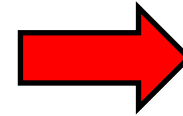
CRISP-DM: Churn Prediction



CRISP-DM: Churn Prediction

- Data Preparation
 - Handle missing values
 - Feature selection
 - Data transformation (IsActiveMember 1:yes, 0:no)

No.		Name
1	☐	RowNumber
2	☐	CustomerId
3	☐	Surname
4	☐	CreditScore
5	☐	Geography
6	☐	Gender
7	☐	Age
8	☐	Tenure
9	☐	Balance
10	☐	NumOfProducts
11	☐	HasCrCard
12	☐	IsActiveMember
13	☐	EstimatedSalary
14	☐	Exited



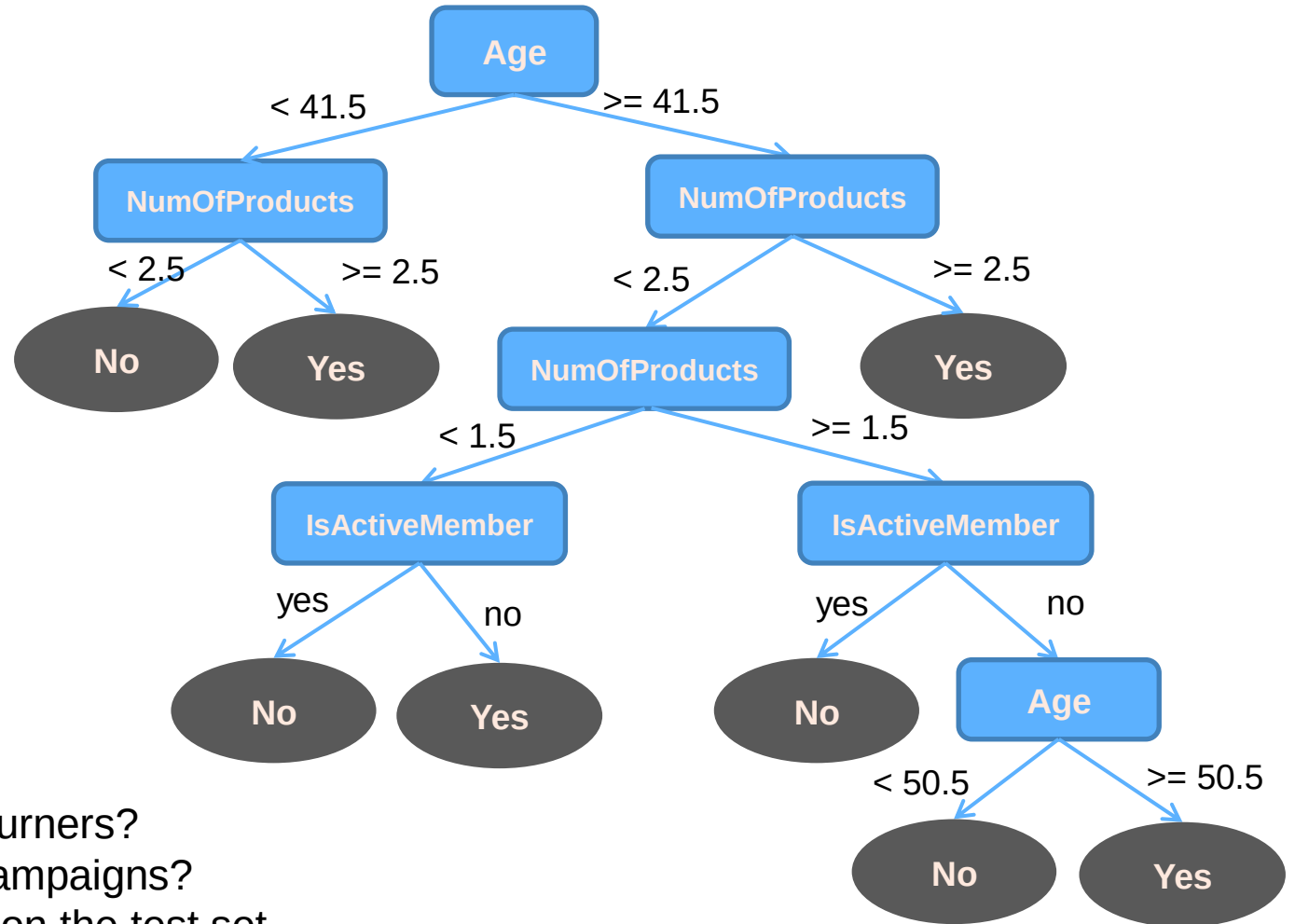
Final dataset includes 6 features

No.		Name
1	☐	Geography
2	☐	Gender
3	☐	Age
4	☐	NumOfProducts
5	☐	IsActiveMember
6	☐	Exited

CRISP-DM: Churn Prediction

- Modeling

- Split train and test dataset
- Decision Tree Model



- Evaluation

- Verify the model meets business and performance goals
- Does the model correctly identify churners?
- Is recall high enough for retention campaigns?
- Recall = 72% and Precision = 68% on the test set
- Even if accuracy is high, recall might be low → the model misses many churners → not acceptable

CRISP-DM: Churn Prediction

- Deployment
 - Integrate the model into business operations
 - Web based application
 - Implement model pipeline
 - Model explanation reports (SHAP)
 - Example:
"Every morning the system assigns a churn probability (0–1) for each customer. Top 5% are sent to the retention team."
- Maintenance (Optional but important)
 - Update model with new data every 3–6 months
 - Track churn rate and campaign response

Enter User Information in the Form Below

Credit Score: 	Number of Products Used: -- select --
Country: -- select --	Has an Active Credit Card: -- select --
Gender: -- select --	Is an Active Bank Member: -- select --
Age: 	Estimated Salary:
Years as Bank Customer: 	
Account Balance: 	Submit

CRISP-DM: Churn Prediction

CRISP-DM Phase	Churn Prediction Focus
Business Understanding	Define churn, business goals, KPIs
Data Understanding	Explore customer behavior and churn patterns
Data Preparation	Clean data, features selection, data transformation
Modeling	Train ML models to classify churn vs. non-churn
Evaluation	Ensure the model delivers business value
Deployment	Integrate predictions into operations and retention
Maintenance (Optional)	Monitor model performance over time

What Kind of Data for Mining?

- Relational databases
- Transactional databases
- Data warehouse
- Data lake
- Data lakehouse
- Advanced DB and information repositories
 - Object-oriented
 - object-relational databases
 - Spatial databases
 - Time-series data and temporal databases
 - Text databases
 - multimedia databases
 - WWW

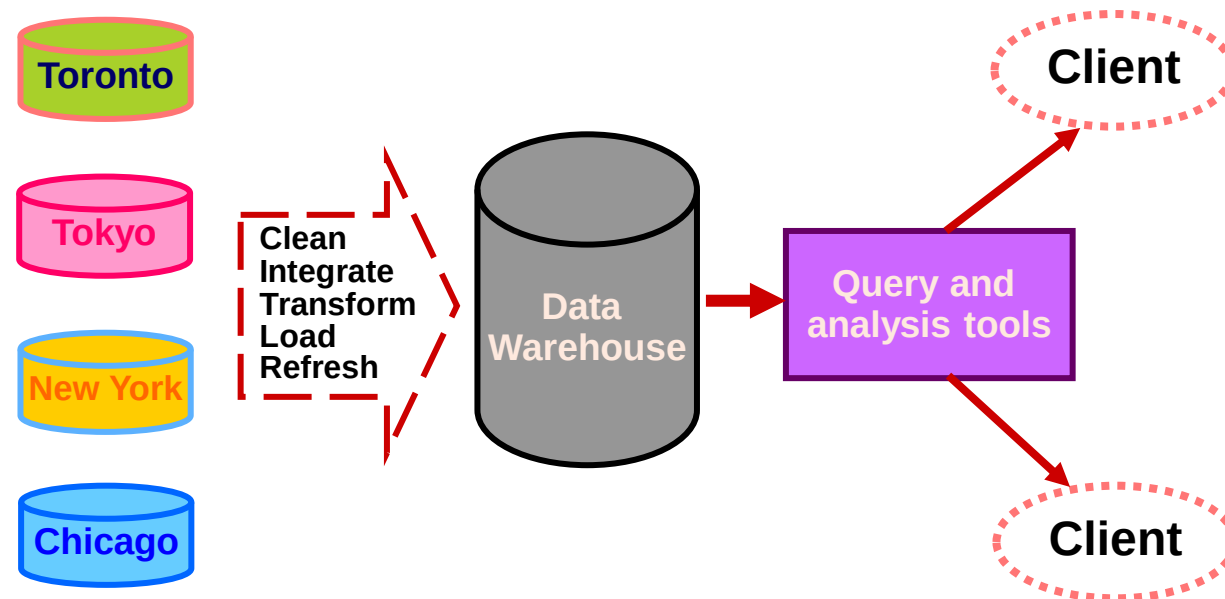
What Kind of Data for Mining?

- Transactional databases
 - Each record represents a transaction
 - A transaction includes a unique transaction identity number (trans_ID) and a list of items making up the transaction

trans_ID	List of item_ID
T100	I1, I3, I8, I16
T200	I2, I8
...	...

What Kind of Data for Mining?

- Data warehouses
 - A repository of information collected from multiple sources, stored under a unified schema



Typical framework of data warehouse

The Problems of Multiple Sources

- Schema Differences
- Naming Differences
- Data Type Differences
- Value Differences
- Semantic Differences
- Missing Values

The Problems of Multiple Sources

The AutoX company has 1,000 branches around the world. Each branch has its own set of databases.

- Schema Differences
 - Branch A: Cars(serialNo, model, color, autoTrans, cdPlayer, ...)
 - Branch B: Autos(serial, model, color)
Options(serial, option)
- Naming Differences
 - Branch A: Table name → Cars
 - Branch B: Table name → Autos
- Data Type Differences
 - Branch A: serialNo → integer
 - Branch B: serial → string

The Problems of Multiple Sources

- Value Differences
 - Branch A: color → “black”
 - Branch B: color → “BL”
- Semantic Differences
 - Branch A: Autos → cars
 - Branch B: Autos → cars and 4x4 W
- Missing Values
 - Branch A: model → Civic DX, LX or EX
 - Branch B: model → Civic

Data Mining Functionality

- Association Analysis
 - Discovery of association rules showing attribute-value conditions that occur frequently together in a given set of data
 - Widely used for market basket analysis
 - single-dimensional association rules
 - `contains(T, "computer") → contains(T, "software")` [support=2%, confidence=75%]
 - multidimensional association rule
 - `age(X, "20..29") and income(X, "20..29K") → buys(X, "PC")` [support = 2%, confidence = 60%]

Data Mining functionality

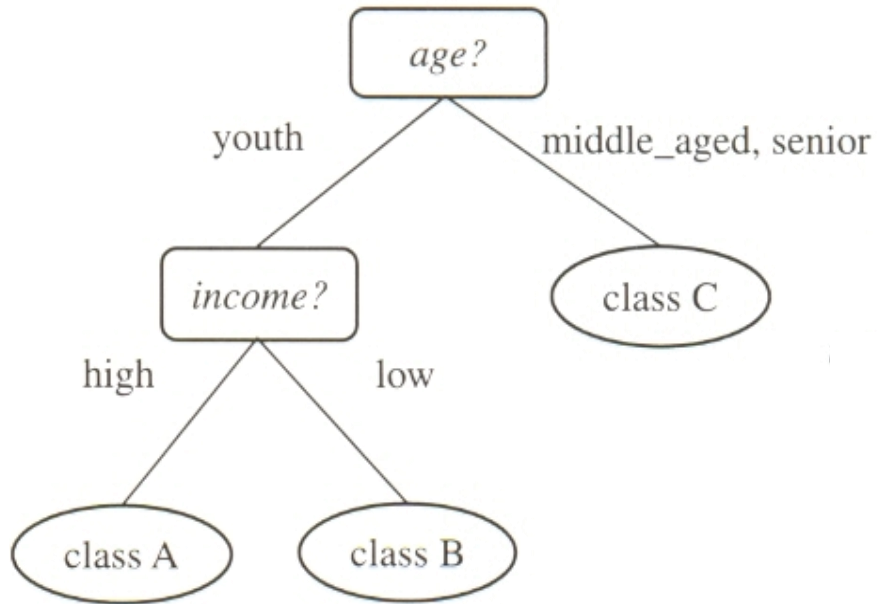
- Classification and Prediction
 - Construct models (functions) based on some training examples
 - Describe and distinguish classes or concepts for future prediction
 - Predict some unknown class labels
 - Typical methods
 - Decision trees, naïve Bayesian, neural networks, rule-based classification

age(X, "youth") AND income(X, "high")	→	class(X, "A")
age(X, "youth") AND income(X, "low")	→	class(X, "B")
age(X, "middle_aged")	→	class(X, "C")
age(X, "senior")	→	class(X, "C")

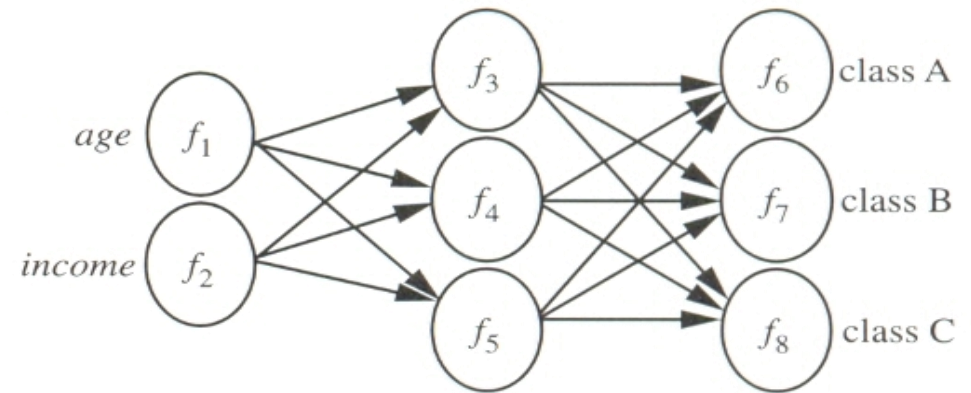
the set of classification rule model

Data Mining functionality

- Classification and Prediction



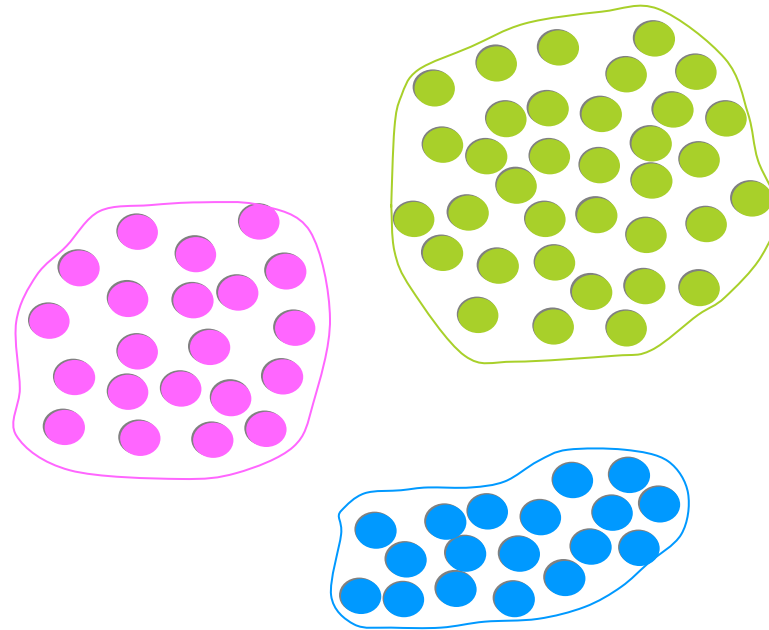
decision tree model



neural network model

Data Mining functionality

- Cluster Analysis
 - Analyze data without consulting a known class label
 - Maximizing the intra-class similarity and minimizing the interclass similarity



Data Science/Data Mining methods



- การจัดกลุ่มข้อมูลตามความคล้ายคลึง (Clustering)



Data Mining functionality

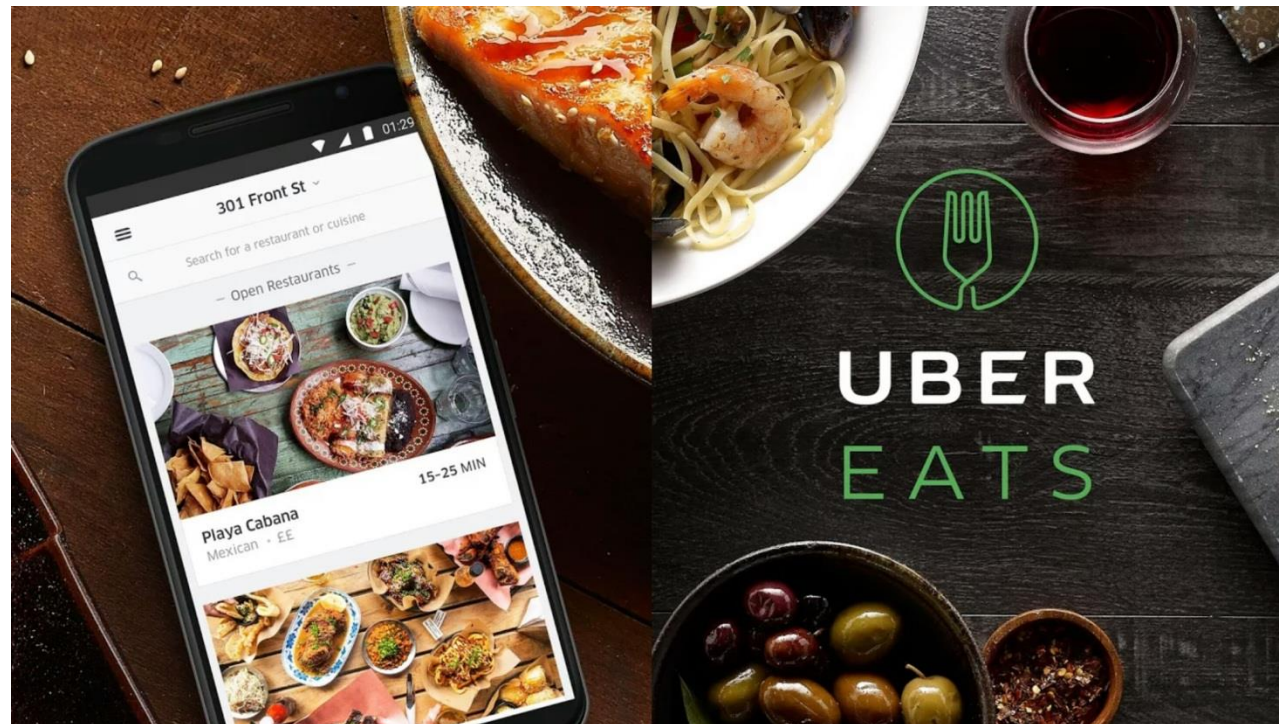
- Outlier Analysis
 - Outlier: a data object that does not comply with the general behavior of the data
 - Noise or exception? – one person's garbage could be another person's treasure
 - Use in fraud detection, rare events analysis
 - Methods: clustering or regression analysis
- Trend and Evolution Analysis
 - Describes and models regularities or trends for objects whose behavior changes over time

Data Mining Applications

- Education
- Commercial
 - Bank: Fraud detection
 - Retail : Market Basket Analysis
- Financial
- Sport
- Entertainment
- Social Network
- Lifestyle
- Medical

Commercial

- Uber Eats
 - Getting hot food delivered quickly
 - Use customer's order history to create virtual restaurant



Commercial

- **McDonald's: Order Prediction**



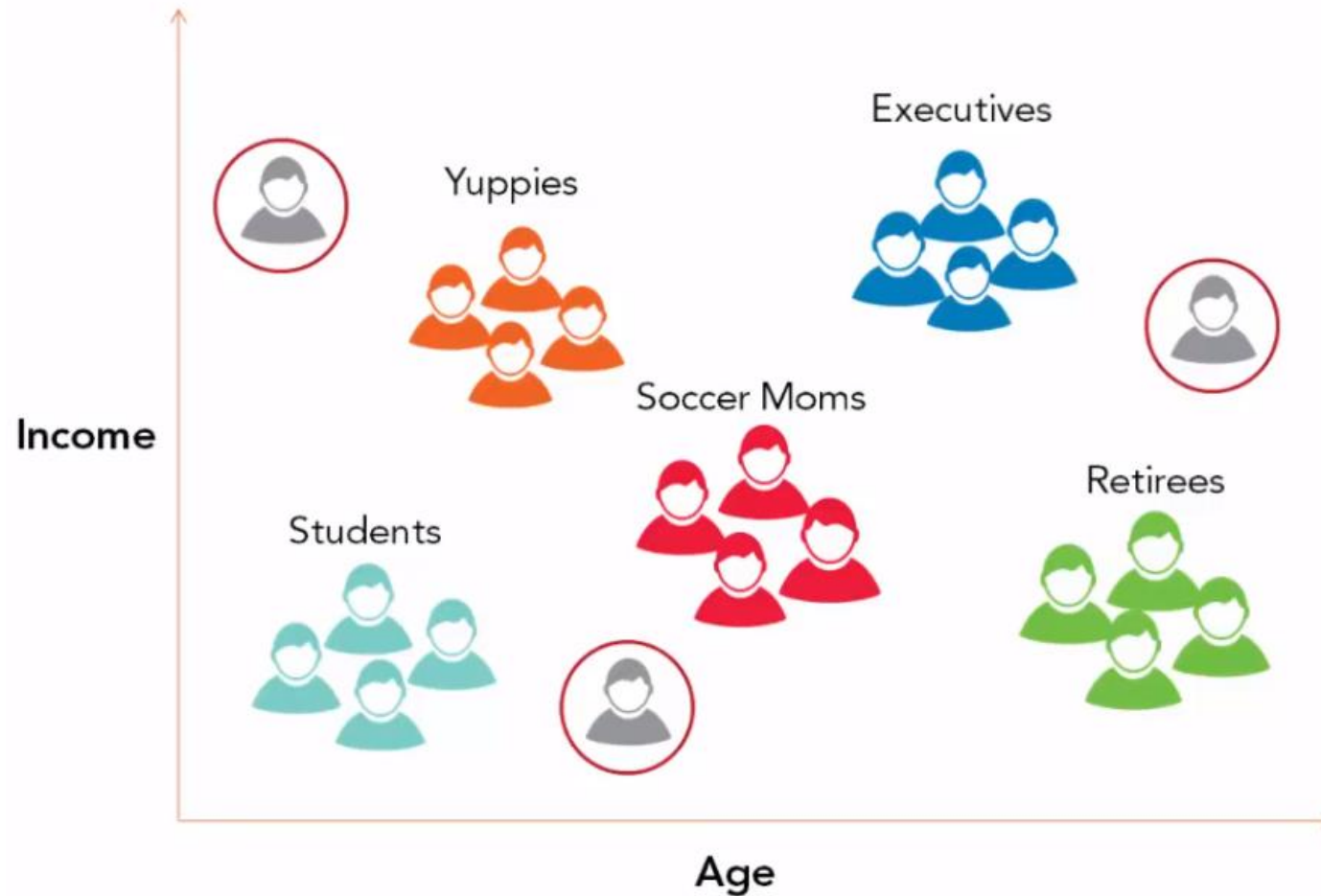
<https://www.bbc.com/news/uk-wales-55811011>

Using **Dynamic Yield's machine-learning** updated the digital drive-thru menu displays based on:

- The time of day
- Current order selections
- Popularity of menu items
- Weather conditions

<https://emerj.com/artificial-intelligence-at-mcdonalds/>

Customer Segmentation



Sentiment Analysis

- Analysis of The Most and Least Expensive London-based Hotels' Review Data



Lifestyle: Netflix

Artwork Personalization



Financial

- Gold price
- Stock market



Medical

- Disease diagnosis
- Effectiveness of treatments

Patient ID	NRTI							NNRTI			PI							II
	ZDV	D4T	DDI	3TC	FTC	ABC	TDF	EFV	NVP	ETV	IDV	SQV/r	NFV	FosAPV/r	LPV/r	ATV/r	TPV/r	
AY030412	R	R	R	R	R	R	R	R	R	G	R	R	I	R	G	G	G	G
AY030413	R	R	G	G	G	G	G	G	G	G	R	I	R	R	I	R	G	G
AY030414	R	R	G	R	R	G	G	G	G	G	G	G	R	G	G	G	G	G
AY030415	R	R	R	G	G	G	I	G	G	G	R	I	R	G	G	R	G	G
AY030420	G	G	G	R	R	G	G	R	R	G	G	G	G	G	G	G	G	G
AY030421	R	R	G	G	G	G	G	G	G	G	G	G	G	G	G	G	G	G
AY030422	R	R	R	G	G	G	G	G	G	G	R	R	R	R	I	G	G	G
AY030426	R	R	G	R	R	G	G	G	G	G	R	R	R	R	G	G	G	G
AY031148	R	R	G	R	R	R	I	G	G	G	R	R	R	R	G	R	G	G

SIR code: S = Susceptible, I = Intermediate, R = Resistant

Algorithm: ANRS Version 2007.09

Levels:

- 1. Susceptible
- 2. Possible resistance
- 3. Resistance

Data Mining Tools

POPULAR DATA MINING TOOLS

IBM SPSS

IBM'S STATISTICAL PACKAGES FOR THE SOCIAL SCIENCES IS A HIT FOR LARGE-SCALE PROJECTS

PYTHON

THE EASE OF USE HAS MADE PYTHON ONE OF THE MOST EFFECTIVE DATA MINING TOOLS

KNIME

KNIME'S USER-FRIENDLY FRAMEWORK FOCUSES ON DATA PIPE-LINING AND INTERACTIVE TABLES

H2O

WITH CUTTING-EDGE TECHNOLOGY, H2O HAS AN ENTHUSIASTIC USER COMMUNITY



R
FREE AND EASY TO PICK UP FOR NON-PROGRAMMING BACKGROUNDS



RAPIDMINER

AN OPEN-SOURCE PREDICTIVE ANALYTICS SOFTWARE FOR DATA MINING PROJECTS



SAS

THE DATA MINING TASKS IN SAS ARE GREAT FOR ENTERPRISE-LEVEL WORK



ORANGE

FREE AND IDEAL FOR BEGINNERS, IT HAS PRE-INSTALLED DATA MINING WORKFLOWS



SPARK

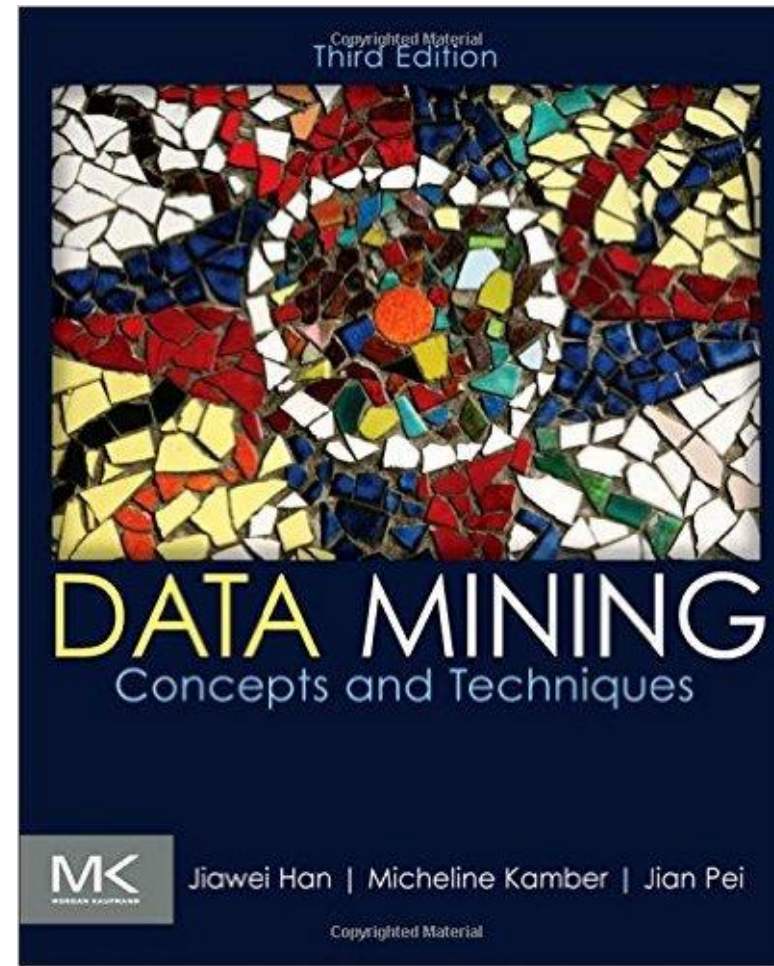
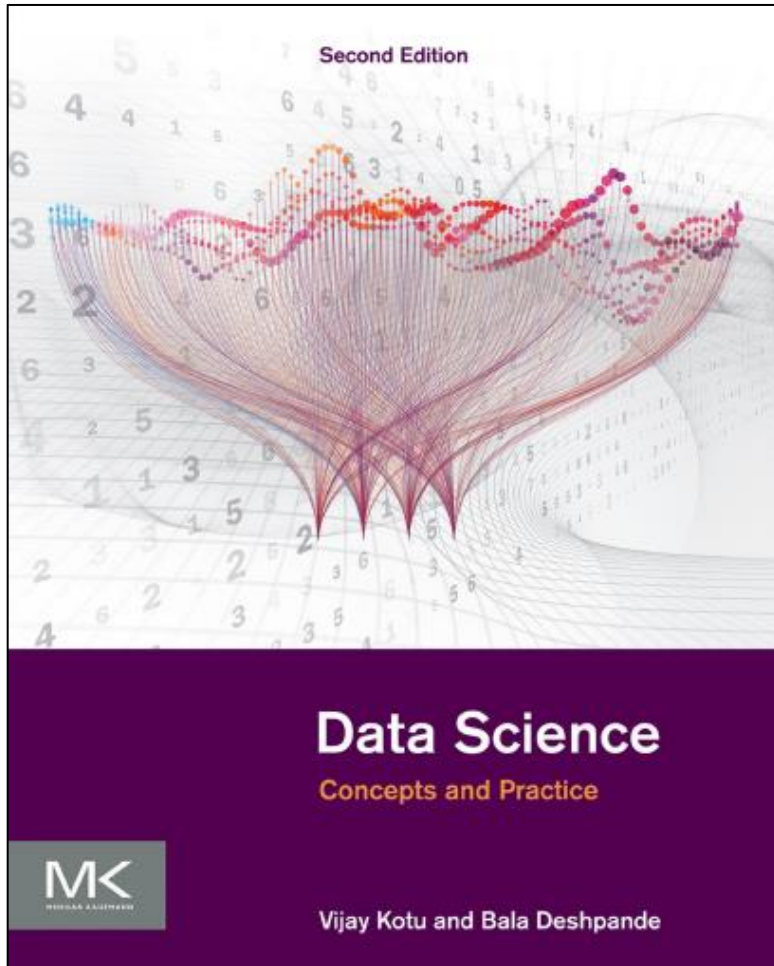
SPARK ALLOWS YOU TO FLOAT THROUGH OCEANS OF COLLECTED DATA WITH EASE



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Textbooks for Data Mining

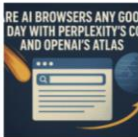


Data Mining Resources

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Are AI Browsers Any Good? A Day with Perplexity's Comet and OpenAI's Atlas

Understanding the underlying technology helps explain why AI browsers exhibit such uneven performance.

By **Shittu Olumide**, Technical Content Specialist on November 26, 2025 in **Artificial Intelligence**



5 Excel AI Lessons I Learned the Hard Way

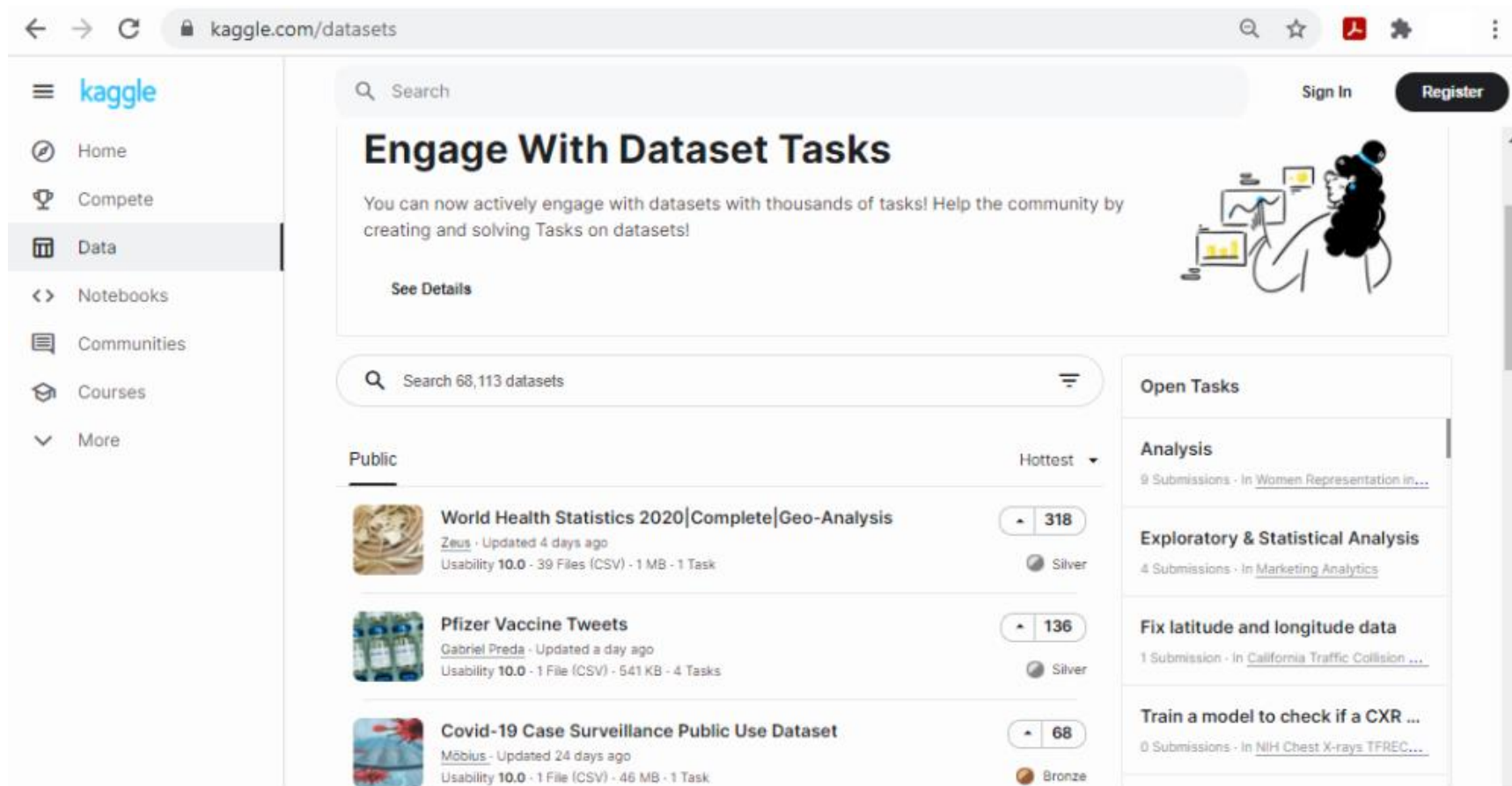
This article transforms the unwelcome experiences into five comprehensive frameworks that will elevate your Excel-based machine learning work.

By **Rachel Kuznetsov**, KDnuggets on November 26, 2025 in **Artificial Intelligence**



Data Mining Resources

- www.kaggle.com: provide datasets and data science competition



The screenshot shows the Kaggle Datasets page. The left sidebar contains navigation links: Home, Compete, Data (selected), Notebooks, Communities, Courses, and More. The main content area features a search bar with the text "Search 68,113 datasets". Below the search bar, there is a section titled "Engage With Dataset Tasks" with a subtext: "You can now actively engage with datasets with thousands of tasks! Help the community by creating and solving Tasks on datasets!" and a "See Details" link. To the right of this section is an illustration of a person interacting with data visualizations. Below the search bar, a list of public datasets is displayed, sorted by "Hottest". The datasets listed are:

- World Health Statistics 2020|Complete|Geo-Analysis** by Zeus, Updated 4 days ago, Usability 10.0, 39 Files (CSV), 1 MB, 1 Task, 318 tasks, Silver medal.
- Pfizer Vaccine Tweets** by Gabriel Preda, Updated a day ago, Usability 10.0, 1 File (CSV), 541 KB, 4 Tasks, 136 tasks, Silver medal.
- Covid-19 Case Surveillance Public Use Dataset** by Möbius, Updated 24 days ago, Usability 10.0, 1 File (CSV), 46 MB, 1 Task, 68 tasks, Bronze medal.

On the right side of the page, there is a section titled "Open Tasks" with a list of tasks:

- Analysis**: 9 Submissions - In [Women Representation in...](#)
- Exploratory & Statistical Analysis**: 4 Submissions - In [Marketing Analytics](#)
- Fix latitude and longitude data**: 1 Submission - In [California Traffic Collision ...](#)
- Train a model to check if a CXR ...**: 0 Submissions - In [NIH Chest X-rays TFREC...](#)

Data Mining Resources

- www.analyticsvidhya.com



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Crash Course to Crack Machine Learning Interview – Part 2: Linear Regression

This is an expert guide on linear regression interview questions along with essential linear regression concepts and terminologies.

Karun Thankachan 07 Nov, 2025



Artificial Intelligence Beginner Machine Learning

How to Run your ML Notebook on Databricks?

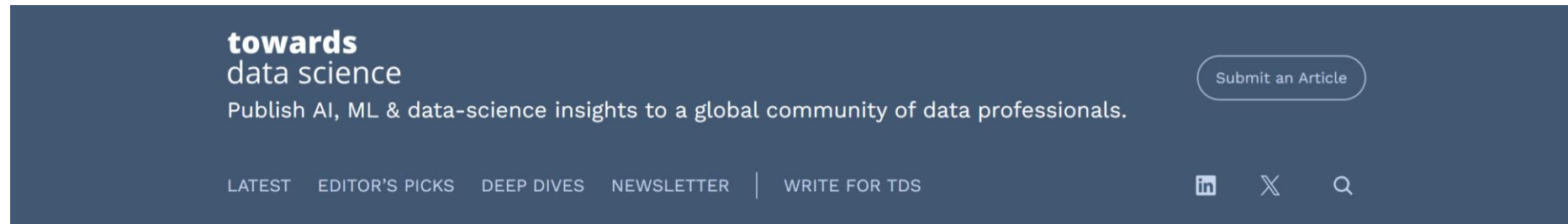
Learn how to run your ML notebook on Databricks with our guide. We'll cover the Free Edition, cluster setup, and experiment tracking.

Janvi Kumari 16 Oct, 2025



Data Mining Resources

- <https://towardsdatascience.com/>



Deep Dives



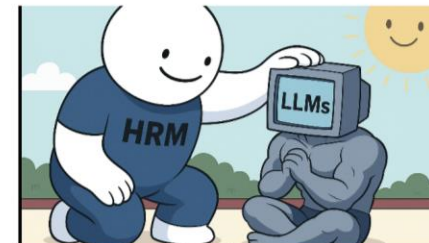
AGENTIC AI

Why CrewAI's Manager-Worker Architecture Fails — and How to Fix It



LLM APPLICATIONS

Ten Lessons of Building LLM Applications for Engineers



ARTIFICIAL INTELLIGENCE

Your Next 'Large' Language Model Might Not Be Large After All